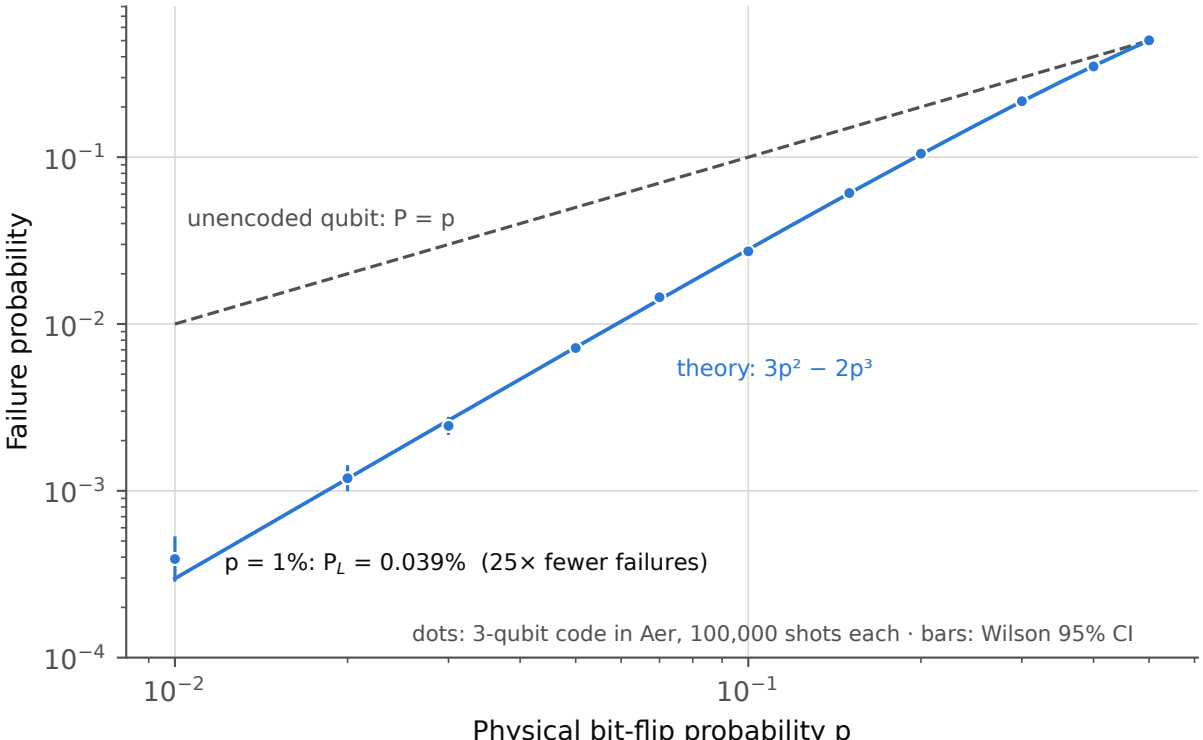
The code turns a 1% bit-flip rate into a 0.04% logical failure rate
Failure probability
10^{-1}
10^{-2}
10^{-3}
10^{-4}
unencoded qubit: P = p
theory: 3p² − 2p³
p = 1%: P_L = 0.039% (25× fewer failures)
dots: 3-qubit code in Aer, 100,000 shots each · bars: Wilson 95% CI
10^{-2}
10^{-1}
Physical bit-flip probability p